$$= \frac{d}{dx}\left(f'(x)\right) = \frac{d}{dx}\left(\frac{d}{dx}f(x)\right)$$

$$= \frac{d^2 f}{dx^2} = \frac{d^2 y}{dx^2}$$

Other notations are $D^2 f(x) = D^2 y = y_2 = y''$

We can interpret a second derivative as a rate of change of a rate of change. But its geometrical interpretation is not so simple. However, there is a close connection exists between the second derivative $f''(x)$ and the radius of curvature of the graph of $y = f(x)$ which you will learn in higher classes.

Similarly, if f'' exists, it might or might not be differentiable. If it is, then the derivative of f'' is called **third derivative** of f and is denoted by

$$f'''(x) = \frac{d^3 y}{dx^3} = y''' = y_3.$$

We can interpret the third derivative physically in case when the function is the position function $f(t)$ of an object that moves along a straight line. Because $s''' = (s'')' = a'(t)$, the third derivative of the position function is the derivative of the acceleration function and is called the **jerk**:

$$j = \frac{da}{dt} = \frac{d^3 s}{dt^3}.$$

Thus, jerk is the rate of change of acceleration.

It is aptly named because a large jerk means a sudden change in acceleration, which causes an abrupt movement in a vehicle.

Example 10.31

Find y', y'' and y''' if $y = x^3 - 6x^2 - 5x + 3$.

Solution

We have, $y = x^3 - 6x^2 - 5x + 3$ and

$$y' = 3x^2 - 12x - 5$$

$$y'' = 6x - 12$$

$$y''' = 6.$$

Example 10.32

Find y''' if $y = \frac{1}{x}$.

Solution

We have, $y = \frac{1}{x} = x^{-1}$

$$y' = -1x^{-2} = -\frac{1}{x^2}$$

$$y'' = (-1)(-2)x^{-3} = \frac{(-1)^2 2!}{x^3}.$$

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